

Zero-Cash Menstrual Health Supply System

Pilot Operating Handbook and Simple App Specification

Version 3.1 — Final Edition (Corrected)

Pilot Model:	One city or village first
Operators:	John in Tanzania and Co-Founder in the USA
Payments:	Direct mobile money APIs (M-Pesa, Airtel, Mixx by Yas, HaloPesa)
API Cost:	No licensing fee; standard telco merchant fees apply (0.5%-2% per transaction)
Telco Fees:	~350 TZS per unit sold (absorbed in price model; no surcharge to customer)
Customer Price:	11,400 TZS (includes telco fee coverage; BoT prohibits surcharges)
Blockchain:	Celo network — audit trail only (no cryptocurrency payments)
Onboarding:	Under 1 minute — phone number auto-maps to blockchain wallet
Transparency:	Every transaction recorded on-chain, publicly verifiable
Fees:	No platform fees; standard mobile money telco fees (0.5%-2%); gas paid by platform
Third Parties:	Only mobile money telcos + Celo (no aggregators, no stablecoin bridges)

This system moves menstrual-health products from approved suppliers to approved local hubs, field champions, and customers using verified mobile-money payments. Every transaction is recorded on a public blockchain for full transparency. The system does not offer credit, loans, automatic deductions, in-app wallets, custody of money, or unlicensed escrow. No stablecoin or cryptocurrency is used for payments. The blockchain serves only as an immutable, publicly verifiable audit trail.

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Executive Summary

This handbook describes a pilot system for distributing menstrual-health products through trusted local entrepreneurs and community health champions in Tanzania. Payments use standard mobile money through direct telco APIs — free to integrate, no third-party fees. Every transaction is recorded on the Celo blockchain as an immutable audit trail, providing full transparency without any cryptocurrency or stablecoin.

Key Design Decisions

Decision	Choice	Reason
Payment method	Direct mobile money APIs (M-Pesa, Airtel, Mixx by Yas, HaloPesa)	Free API access, no licensing fees, no aggregator margin, already familiar to all Tanzanians
Stablecoin	REMOVED	Adds complexity, depends on third-party bridge providers, regulatory uncertainty, not sustainable long-term
Blockchain	Celo — audit trail only	Records events immutably; does not handle money; phone-number mapping enables 1-minute onboarding
Onboarding	Under 1 minute via Celo SocialConnect	Phone number auto-maps to wallet address; no wallet setup, no private keys, no cryptocurrency
Smart contracts	2 contracts (AuditTrail + CustodyChain)	Simplified from 5 — no payment settlement on-chain, only event recording
Dependencies	Mobile money telcos + Celo	No stablecoin bridge, no aggregator, no cryptocurrency exchange

Mobile Money Providers Supported

Provider	Parent Company	Market Share	API Cost	USSD Code
M-Pesa	Vodacom (Vodafone)	~41%	Free	*150*00#
Mixx by Yas	MIC Tanzania (ex-Tigo)	~30%	Free	*150*01#
Airtel Money	Airtel Tanzania	~17%	Free	*150*60#
HaloPesa	Halotel	~9%	Free	*150*88#
T-Pesa	TTCL	~1%	Free	*150*71#
Azam Pesa	Azam	~0.1%	Free	—

All provider APIs have no licensing fee and no monthly subscription. However, each telco charges merchant fees on every transaction: M-Pesa charges 0.5%-1.5%, Airtel charges 1.5%-2%.¹ These fees are the cost of using mobile money infrastructure and cannot be eliminated. The customer price is set at 11,400 TZS to cover all telco fees while preserving stakeholder margins. The Bank of Tanzania prohibits merchants from imposing surcharges on digital transactions, so fees must be absorbed into the price model, not added as a separate charge.²

Celo is a mobile-first blockchain with built-in SocialConnect, a privacy-first naming service that maps phone numbers to wallet addresses automatically.³ This enables onboarding in under 1 minute without any wallet management or cryptocurrency knowledge.

1. McTaba, "Accept Mobile Money Payments on Website Tanzania."
<https://www.mctaba.com/learn/tanzania/accept-mobile-money-payments-website-tanzania>
2. McTaba, "Selcom vs ClickPesa vs Pesapal vs Direct API Tanzania."
<https://www.mctaba.com/learn/tanzania/selcom-vs-clickpesa-vs-pesapal-vs-direct-api-tanzania>
3. Binance, "Celo Talks: Mobile Payments and Web3" — SocialConnect maps wallets to phone numbers. <https://www.binance.com/en/square/post/08-05-2025-binance-interviews-from-layer-1-to-ethereum-layer-2-celo-talks-celo-token-mobile-payments-and-the-future-of-web3-27771720218849>

1. Purpose and Rules

Purpose

The system helps make menstrual-health products more available through trusted local entrepreneurs and community health champions. All payments use standard mobile money (M-Pesa, Airtel, Mixx by Yas, HaloPesa) through direct API integration. Every transaction is recorded on the Celo blockchain as an immutable audit trail, providing full transparency for all stakeholders without any cryptocurrency or stablecoin.

It supports:

- Approved supplier product sourcing with on-chain batch tracking.
- Verified mobile-money payment confirmation via direct telco API.
- Product batch and inventory tracking with on-chain custody events.
- Clear ownership transfer recorded immutably on the Celo blockchain.
- Simple voluntary customer installment payments with on-chain progress tracking.
- Digital receipts and transparent margins visible on the blockchain explorer.
- Safe distribution of reusable and disposable menstrual products.
- Real-time reconciliation through automated on-chain event matching.

Non-negotiable Rules

No credit

- A customer does not receive a product before completing payment unless the product is explicitly funded by a documented donor, subsidy, or approved sponsor.
- The app never creates a negative customer balance.
- The app never auto-debits a customer, champion, rider, or hub.
- Smart contracts enforce this rule: no product-release event can be recorded without a confirmed payment.

No self-signup

- Only John and the USA Co-Founder can add, approve, suspend, or remove system participants.
- A phone number alone is not enough. Each participant must be personally verified before activation.
- Each approved stakeholder's phone number is auto-mapped to a Celo wallet address via SocialConnect — no wallet setup required.

No cash handled by the platform

- Payments occur through standard mobile money (M-Pesa, Airtel, Mixx by Yas, HaloPesa) via direct telco APIs.
- The app receives payment confirmation directly from the telco's API webhook.
- The app records the confirmed payment as an audit event on the Celo blockchain.
- The platform does not hold participant money, custody stablecoin, or act as a bank, wallet, or escrow provider.
- No stablecoin, cryptocurrency, or digital asset is used for any payment.

No payment confirmation from screenshots or SMS alone

- The app recognizes payment only after the mobile money provider's API confirms the transaction.
- A customer-facing 'payment sent' screen means pending until the telco API webhook confirms.
- The blockchain audit record is written after confirmation, providing additional immutable proof.

No hidden deductions

- All margin distributions are calculated by the app and recorded as audit events on-chain.
- Inventory shortages, damaged stock, disputes, returns, or suspected theft are reviewed manually by John and the USA Co-Founder.
- The system does not automatically take money from any participant's mobile-money wallet.

No unnecessary personal data

- Collect only what is needed: name, verified phone number, role, service area, and transaction records.
- Do not collect menstrual-cycle details, diagnoses, or precise customer location unless strictly necessary and lawful.
- Phone numbers are hashed before recording on the Celo blockchain; personal data is stored off-chain.
- Tanzania's PDPC states that personal data must not be collected or processed without registration as a data controller or processor.¹

No reuse without safe conditions

- Reusable products are offered only where a participant has access to adequate water, soap, privacy, washing space, and drying conditions.
- Reusable menstrual materials must be washed and dried after use; this must be taught clearly before sale or distribution.²

No expansion before proof

- Start with one supplier route, one city or village, one hub, one or two riders, and a small group of trained field champions.
- Do not add more villages until on-chain reconciliation shows payments, inventory, customer service, and reporting are working consistently.
- All pilot metrics are verifiable on the Celo blockchain explorer, providing objective evidence of readiness for expansion.

1. Tanzania PDPC. <https://pdpc.go.tz/>
2. UNICEF, Menstrual Hygiene Products. <https://www.unicef.org/supply/menstrual-hygiene-products>

2. Mobile Money Payment Integration

The system accepts payments from all major Tanzanian mobile money providers through direct API integration. No aggregator, no stablecoin bridge, no third-party payment processor. The app connects directly to each telco's API, receives payment confirmation webhooks, and records the event on the Celo blockchain.

2.1 Supported Providers

Provider	Parent	Share	API Portal	USSD	Status
M-Pesa	Vodacom	~41%	business.m-pesa.com	*150*00#	Phase 1 (launch)
Mixx by Yas	MIC Tanzania	~30%	tigo.co.tz/developers	*150*01#	Phase 2
Airtel Money	Airtel	~17%	developers.airtel.co.tz	*150*60#	Phase 2
HaloPesa	Halotel	~9%	halotel.co.tz/halopesa	*150*88#	Phase 3
T-Pesa	TTCL	~1%	ttcl.co.tz/t-pesa	*150*71#	Phase 3
Azam Pesa	Azam	~0.1%	azampay.com	—	Phase 3

Phase 1 launches with M-Pesa only, covering ~41% of the Tanzanian market. M-Pesa's API (Daraja) is the most mature, best documented, and completely free — no licensing fee, no monthly subscription, no per-transaction API charge.¹ Phase 2 adds Airtel Money and Mixx by Yas, covering an additional ~47% (total ~88%). Phase 3 adds the remaining providers.

2.2 API Cost Structure

Cost Type	Direct API (This System)	Aggregator (Eliminated)
Developer account	No licensing fee	No licensing fee
API access	Free to register and test	Free to register
Monthly subscription	None	None
Per-transaction fee	0.5%-2% telco merchant fee	Telco fee + 1-3% aggregator margin
Platform fee	Zero	Zero
Stablecoin conversion	N/A — no stablecoin	N/A — no stablecoin
Gas (blockchain)	Sub-cent, paid by platform budget	Sub-cent, paid by platform budget
Total per unit	~350 TZS (telco fees only)	~500-600 TZS (telco + aggregator)

Direct API integration avoids the 1-3% aggregator markup that would be added on top of telco fees. The irreducible cost is the telco merchant fee (0.5%-2% per transaction), which is the cost of using mobile money infrastructure in Tanzania. This cannot be eliminated without removing mobile money itself. The customer price of 11,400 TZS covers all telco fees while preserving stated stakeholder margins.

2.3 Integration Requirements

- Registered Tanzanian business: Required by each telco for commercial API access.

- Commercial approval: Each telco's business team must approve the merchant account.
- Compliance review: Standard KYC/AML checks performed by the telco.
- API keys: Generated from each telco's developer portal (free).
- Webhook endpoint: The app provides a URL where each telco sends payment confirmation notifications.
- Sandbox testing: Each telco provides a free sandbox environment for testing before going live.

2.4 Payment Flow

When a customer makes a payment, the flow is:

Step	Action	What Happens
1	Customer dials USSD or uses mobile money app to send payment	Standard mobile money transaction initiated
2	Telco processes payment and sends webhook to app backend	Payment confirmation received via API
3	App backend verifies webhook authenticity: (a) check webhook signature, (b) call telco API directly to confirm transaction (Transaction Status Query), (c) check for duplicate transaction reference	Payment confirmed as genuine — blocks spoofing, replay, and duplicate attacks
4	App backend writes AuditEvent to Celo blockchain	Immutable on-chain record created
5	App updates status to PAYMENT_CONFIRMED	Status change visible in app
6	App generates receipt with blockchain explorer link	Customer receives receipt with verification link

Total time from payment to confirmation: typically 5-15 seconds (telco API response + blockchain audit write). The customer does not wait for blockchain confirmation — the telco API confirmation is the payment confirmation. The blockchain record is written afterwards as proof.

2.5 Offline Handling

If the telco API is slow or the app backend loses connectivity, the app shows 'Payment pending — verifying with provider.' No product is released until confirmation is received. If the Celo blockchain is temporarily unavailable, payments still work normally — audit events are queued locally and written to the blockchain when connectivity returns. The app shows 'Audit trail: syncing' status.

2.6 Regulatory Compliance

Payment services in Tanzania are regulated by the Bank of Tanzania under the National Payment Systems Act 2015.² Mobile money providers (Vodacom, Airtel, MIC Tanzania, Halotel) are licensed and regulated by both the Bank of Tanzania and the Tanzania Communications Regulatory Authority (TCRA).³ By using direct telco APIs, the system operates entirely within the existing regulatory framework — no cryptocurrency regulation, no stablecoin sandbox, no new licensing required.

1. ClickUp Tanzania, "Vodacom provides Daraja API access to registered businesses at no API licensing cost." <https://www.clickup.co.tz/service-fintech.html>
2. Bank of Tanzania, National Payment Systems Annual Report. <https://www.bot.go.tz/Publications/Regular/Annual%20Report/en/2024082116493467.pdf>
3. Payments.gi, "Tanzania Payments Regulation Monitor." <https://payments.gi/jurisdictions/tanzania/>

3. Blockchain Transparency Layer (Audit Trail)

The blockchain is a transparency ledger, not a payment system. It records events that already happened. It does not move money, settle payments, or distribute margins. Those happen through standard mobile money and the app backend. The blockchain's job is to make those events publicly verifiable and immutable.

3.1 Why Celo

Celo is a mobile-first blockchain specifically designed for emerging markets. It was chosen for three reasons:

- Phone-number mapping: Celo's SocialConnect protocol maps phone numbers to wallet addresses automatically, enabling 1-minute onboarding without wallet setup or private key management.¹
- Near-zero gas fees: Transaction costs on Celo are sub-cent, making it affordable to record every event on-chain.
- Mobile-optimized: Celo is designed for smartphones and works well in low-bandwidth environments common in rural Tanzania.

3.2 What Gets Recorded On-Chain

Event	When	What is recorded
PaymentConfirmed	Mobile money payment verified via telco API	Amount, currency, batch/order ID, phone hash, provider, timestamp
CustodyTransferred	Stock moves between stakeholders	From phone hash, to phone hash, batch ID, quantity, custody state, timestamp
MarginDistributed	App calculates margin after sale	Stakeholder phone hash, role, amount, linked sale, timestamp
HandoverCompleted	Customer receives product	Customer phone hash, product type, batch ID, champion phone hash, timestamp
ExceptionRaised	Problem reported	Exception type, batch ID, reporter phone hash, timestamp
ExceptionResolved	Admin resolves problem	Resolution, admin phone hash, timestamp
StakeholderActivated	New stakeholder approved	Phone hash, role, service area hash, timestamp
PriceListUpdated	Admin changes price (dual-signature)	Product, old price, new price, admin signatures, timestamp
DailyReconciliation	Automated daily check	Total events, matched count, mismatched count, timestamp

3.3 Onboarding — Under 1 Minute

When the admin enters a stakeholder's verified phone number, the app backend uses Celo's SocialConnect to create or link a wallet address to that phone number. The stakeholder never sees a wallet, private key, or cryptocurrency. They just tap a link, enter their phone number, create a 4-digit PIN, and start working.

Step	What admin does	What stakeholder does	Time
1	Enters phone number, role, service area in dashboard	Nothing — waiting	10 sec
2	App calls Celo SocialConnect to map phone to wallet	Nothing — automatic	5 sec
3	App sends SMS/WhatsApp with activation link	Receives SMS	10 sec
4	App assigns role-specific first task	Opens link, enters phone, creates PIN	15 sec
5	Wallet active on Celo — ready to record events	Sees first task — ready to work	5 sec
	TOTAL		~45 sec

What the stakeholder does NOT see:

- No wallet creation screen
- No private key, seed phrase, or recovery words
- No cryptocurrency purchase or conversion
- No gas fee payment
- No blockchain technical details
- No stablecoin, token, or digital asset

3.4 Transparency for All Stakeholders

Each stakeholder can verify their own transactions on a public blockchain explorer (e.g., celoscan.io) without trusting the app or any individual. The app provides a 'View on Blockchain' link for every transaction.

Stakeholder	What they can verify on-chain
Customer	Payment was received; product handover was completed
Champion	Margin payments; handover confirmations
Hub Manager	Incoming stock custody; outgoing champion transfers
Rider	Pickup confirmations; delivery confirmations; margin payments
Supplier	Batch readiness confirmations; payment receipts
Admins	Full audit trail of all events, margins, exceptions, resolutions
USA Co-Founder	Cross-border verification of all transactions without bank access

3.5 Privacy on the Blockchain

No personal data is stored on the public blockchain. Phone numbers are hashed (converted to an unrecognizable string) before recording. Only the following appears on-chain:

- Phone number hashes (not actual numbers)
- Transaction amounts and currency

- Batch IDs and custody states
- Event types and timestamps
- No names, addresses, health data, or customer-identifying information

3.6 What Happens When the Blockchain Is Down

- Mobile money payments continue to work normally (telco API is independent of blockchain).
- The app shows 'Audit trail: syncing' instead of 'Audit trail: verified'.
- Audit events are queued locally and written to Celo when connectivity returns.
- No payment is delayed, blocked, or failed due to blockchain downtime.
- This is fundamentally different from a stablecoin-based system, where blockchain downtime means payment failure.

3.7 Blockchain Explorer Access

Every transaction in the app includes a 'View on Blockchain' button that opens the transaction on celoscan.io. The app shows wallet addresses in simplified format (e.g., 'Hub-7A3F...2B9C') rather than raw hex, making it readable while remaining verifiable.

3.8 Cost

Total monthly blockchain cost for the pilot scale is under \$1 USD in gas fees, paid by the platform budget. No stakeholder pays any blockchain fee. No stablecoin, no bridge fees, no conversion charges.

1. CoinGecko, "What is Celo?" — Celo links phone numbers to public keys; any mobile number functions as a wallet.
<https://www.coingecko.com/learn/celo>

4. Roles

Each approved stakeholder's phone number is auto-mapped to a Celo wallet address via SocialConnect. Stakeholders do not manage wallets or cryptocurrency — the app handles all blockchain interactions automatically.

Role	Who adds	Main responsibility	Blockchain interaction
Super Admin	N/A	John and USA Co-Founder jointly govern	Approve on-chain price updates, resolve exceptions, view full audit trail
Supplier / Factory	After business verification	Prepare approved batches for pickup	Confirm batch readiness on-chain, receive payment via mobile money
Boss Rider	After identity and training verification	Buy stock, transport to hub	Confirm pickup on-chain, receive margin via mobile money
Hub Manager	After site and training verification	Inspect, receive, store, sell stock	Confirm receipt on-chain, manage stock custody
Field Champion	After trust and training verification	Educate, sell, record payments, hand over	Enroll customer, record payments on-chain, complete handover
Customer	By verified champion after consent	Make payments, receive product	View on-chain payment progress, receive receipt with blockchain link

Pilot Stakeholder Standard

Before adding any person, John and the USA Co-Founder confirm:

- Full name.
- Verified phone number (auto-mapped to Celo wallet via SocialConnect).
- Correct role.
- Community location or service area.
- Mobile-money account (M-Pesa, Airtel, Mixx by Yas, or HaloPesa).
- Training completed.
- Agreement to follow the operating rules.
- Emergency and escalation contact.
- No unresolved history of fraud, serious misconduct, or unsafe conduct.

For the first pilot, choose people who are already known and trusted in the community.

5. Simple Money Model

The first pilot uses one approved product price list. Prices are stored on-chain in a smart contract, ensuring no participant can alter prices without dual-admin approval. The customer price is set at 11,400 TZS to cover standard telco merchant fees (approximately 350 TZS per unit across four transactions) while preserving the stated stakeholder margins. The Bank of Tanzania prohibits merchants from imposing surcharges on digital transactions, so telco fees must be absorbed into the price model.¹

Example only: standard kit

Step	Buyer	Seller	Unit price	Seller margin
Factory sale	Boss Rider	Supplier / Factory	7,500 TZS	Supplier's agreed price includes cost and margin
Hub sale	Hub Manager	Boss Rider	8,000 TZS	Boss Rider earns 500 TZS (mobile money, recorded on-chain)
Champion sale	Field Champion	Hub Manager	9,000 TZS	Hub Manager earns 1,000 TZS (mobile money, recorded on-chain)
Customer sale	Customer	Field Champion	11,400 TZS	Champion earns 2,000 TZS + 400 TZS fee coverage (mobile money, on-chain)
Telco fees	Absorbed in price	All parties	~350 TZS	0.5%-2% per transaction across 4 transactions; no surcharge to customer

Required Price Rules

- Every price list is approved by both John and the USA Co-Founder and stored on-chain.
- Every price list has a start date, product name, product category, supplier, and region.
- The app shows the full customer price before the customer begins installments.
- All mobile-money charges must be identified in advance: who pays the fee, what the fee is.
- No participant may increase the customer price without written approval recorded on-chain.
- No participant may promise a discount, credit, donation, or free product unless approved in the app.
- Smart contracts enforce exact price amounts; no manual override is possible.

What the App Does Not Do

- Hold customer money or rider money.
- Create in-app wallet balances.
- Collect payment automatically or 'pull' money from any phone.
- Create debt, charge late fees, or give loans.
- Release money based on screenshots or SMS forwards.
- Use stablecoin, cryptocurrency, or digital assets for any payment.

- Custody any funds on behalf of users.

Payment services in Tanzania are regulated by the Bank of Tanzania. This pilot uses licensed mobile money providers via direct API integration rather than operating its own payment system.¹

1. Bank of Tanzania, Payment Systems Annual Report 2024.

<https://www.bot.go.tz/Publications/Regular/Annual%20Report/sw/2025032515311662.pdf>

6. Stakeholder Approval Process

When a stakeholder is activated, the app automatically generates a Celo wallet address linked to their phone number via SocialConnect. The stakeholder does not manage private keys — the app uses SocialConnect's identity layer. The Admin controls activation, suspension, and removal.

Step 1: Person contacts John or USA Co-Founder

The person does not receive an app account at this point.

Next action: Review Stakeholder Request

Step 2: Admin verification

John and the USA Co-Founder jointly verify identity, phone-number control, community reputation, role suitability, mobile-money readiness, storage conditions (hubs), transport reliability (riders), and training (champions).

Next action: Approve, Decline, or Request More Information

Step 3: Training and role agreement

Before activation, each stakeholder receives role-specific training and confirms:

- "I understand my role."
- "I will use only approved prices (enforced on-chain)."
- "I will not sell on unapproved credit."
- "I will not share customer information."
- "I will report damaged, missing, or disputed stock immediately."
- "I understand that all transactions need confirmed mobile money payment before completion."

Next action: Mark Training Complete

Step 4: Activation (under 1 minute)

The Admin enters the stakeholder's verified phone number, role, and service area. The app calls Celo SocialConnect to map the phone number to a wallet address. A StakeholderActivated event is recorded on-chain. The stakeholder receives an SMS with an activation link, enters their phone number, creates a 4-digit PIN, and sees their first task.

Next action: Confirm My Phone and Create My PIN

Step 5: First-day check

Within the first operating day, John confirms the stakeholder can: open the app, read their next action, confirm a test notification, contact support, understand pending status, and view their wallet address on the blockchain explorer.

Next action: Complete First-Day Check

7. App Design Rules

The app must be extremely simple.

One Screen Rule

Each user sees: their role, current status, one large next-action button, a short explanation in English and Swahili, a 'View on Blockchain' link for transactions, and a help button.

Example screen:

```
YOUR STATUS:
Waiting for payment confirmation

NEXT ACTION:
Do not hand over the product yet.

PAYMENT:
Customer paid via M-Pesa – verifying with provider

BUTTON:
Refresh payment status

HELP:
Call John / Send WhatsApp
```

Every screen must show:

- Current role, current status, one next action.
- Transaction or batch reference number.
- On-chain transaction hash (when available, as 'View on Blockchain' link).
- Time of last update, help button, logout button.
- Language option: English / Swahili.

The app must never show:

- Complex menus or raw blockchain technical details (hex, gas, nonce).
- Other users' private information or customer health information.
- 'Paid' status before telco API confirmation.
- Private keys, seed phrases, or wallet management interfaces.
- Multiple conflicting action buttons.

Blockchain Information Display

Blockchain concept	What app shows	What user sees
Transaction hash	'View on Blockchain' button	Opens explorer link
Wallet address	Role + short ID (e.g., 'Hub-7A3F')	Simplified label
Block confirmation	'Payment confirmed' status	Green checkmark
Audit event	'Payment recorded on blockchain'	Plain status text

Blockchain concept	What app shows	What user sees
Gas fee	Not shown to user	Absorbed by platform budget
Margin record	'Your margin: 2,000 TZS received'	Amount + 'verified' badge

Security Controls (from Red Team Analysis)

The following security controls are mandatory and non-negotiable:

- Webhook verification: Every payment webhook is verified by (a) signature check, (b) direct API callback to telco, (c) duplicate transaction reference check. No webhook is accepted without all three checks.
- Rate limiting: Maximum 10 webhooks per phone number per minute. Exceeding this triggers an alert and temporary block.
- IP allowlisting: Webhook endpoint only accepts requests from known telco IP ranges.
- PIN security: 4-digit PIN, maximum 5 attempts, 30-minute lockout after 3 failures, admin reactivation required.
- Session timeout: Automatic logout after 15 minutes of inactivity.
- Key management: App backend wallet private key stored in HSM or secure enclave — never in environment variables.
- Multi-signature admin: All admin actions (price changes, exception resolution) require 2-of-2 dual-signature (John + USA Co-Founder).
- Emergency pause: Admins can immediately halt all smart contract operations if compromise is detected.
- Customer phone verification: SMS verification of customer phone number before enrollment is complete.
- Append-only offline queue: Offline events are persisted to disk, append-only, with tamper-evident hashing of webhook payloads.
- Admin action logging: All admin dashboard actions logged with admin identity, timestamp, and action details.
- Receipt integrity: All receipts generated server-side; champions cannot edit receipt content or blockchain explorer URLs.

8. Core Workflows

A. Supplier / Factory Workflow

Purpose: Prepare product batch for an approved rider. Batch readiness confirmed on-chain.

Status	What supplier sees	Required action	Next app action
No pickup assigned	'No approved pickup scheduled'	Wait	View Upcoming Pickups
Pickup assigned	Product type, unit count, pickup date, rider name	Prepare approved quantity	Confirm Batch Ready
Batch ready	'Batch ready for rider payment and pickup'	Wait for confirmed payment	View Payment Status
Payment pending	'Payment being verified with provider'	Do not release goods	Refresh Payment Status
Payment confirmed	'Payment confirmed. Prepare pickup verification.' (AuditEvent on-chain)	Verify quantity and batch IDs	Confirm Pickup
Pickup complete	'Stock has left factory' (CustodyTransferred on-chain)	Keep digital record	View Completed Pickup
Problem reported	'A problem was reported'	Select reason	Report Exception

Factory release rule: The supplier releases stock only after the mobile money provider's API confirms payment and the rider is verified in the app. The smart contract prevents the CustodyTransferred event without a matching PaymentConfirmed event.

B. Boss Rider Workflow

Purpose: Purchase stock, collect it, transport it safely, and sell it to the assigned hub.

Status	What rider sees	Required action	Next app action
No assignment	'No pickup assigned to you'	Wait	Refresh
Pickup available	Pickup location, product, quantity, total cost	Review and accept	Accept Pickup
Awaiting payment	'Pay supplier via approved mobile money'	Make payment (M-Pesa/Airtel/Mixx/HaloPesa)	I Have Paid
Payment pending	'Payment being verified with provider. Do not collect yet.'	Wait	Refresh Payment Status
Payment confirmed	'You may collect the batch' (PaymentConfirmed on-chain)	Travel to supplier; inspect quantity and seal	Confirm Pickup
In transit	Hub name and delivery reference	Transport directly to assigned hub	Open Delivery Code
At hub	'Show delivery code to hub manager'	Present product and code; wait for inspection	Confirm Delivery Inspection

Status	What rider sees	Required action	Next app action
Hub inspection pending	'Hub is counting and checking stock'	Wait	Refresh Delivery Status
Hub accepted	'Hub accepted stock. Await payment confirmation.'	Do not leave until status clear	Refresh Payment Status
Hub payment confirmed	'Sale complete. Your margin sent via mobile money.' (MarginDistributed on-chain)	Keep digital receipt with blockchain link	View Completed Delivery
Delivery problem	'Do not transfer stock or leave the site'	Contact support	Report Problem

Rider Rules

- The rider carries only approved product quantities.
- The rider does not change the destination hub without Admin approval.
- The rider does not hand stock to an unapproved person.
- The rider does not accept screenshots or verbal promises as payment proof — only telco API confirmation counts.
- If stock is damaged, wet, lost, stolen, or tampered with, the rider reports immediately (ExceptionRaised on-chain).

C. Hub Manager Workflow

Purpose: Receive, inspect, store, and sell inventory to approved field champions.

Status	What hub sees	Required action	Next app action
No incoming delivery	'No rider delivery expected now'	Wait	View Inventory
Rider arriving	Rider reference, expected product, expected quantity	Prepare inspection space	Start Inspection
Inspection required	'Check seal, count units, inspect condition'	Inspect physical stock	Accept / Report Shortage / Report Damage
Stock accepted	'Confirm payment to rider via mobile money'	Make approved payment	I Have Paid
Rider payment pending	'Payment being verified with provider'	Do not mark transfer complete	Refresh Payment Status
Rider payment confirmed	'Stock recorded at your hub' (CustodyTransferred on-chain)	Store stock safely	View Hub Inventory
Champion request	Champion name, product type, unit count, approved price	Confirm available stock	Prepare Champion Transfer
Champion payment pending	'Champion payment being verified'	Do not release stock	Refresh Payment Status
Champion payment confirmed	'Release approved quantity to champion'	Count stock with champion and confirm	Confirm Champion Handover

Status	What hub sees	Required action	Next app action
Low stock	'Stock below minimum level'	Request replenishment	Request Restock
Product safety alert	'Review product category and WASH conditions'	Follow approved guidance	Confirm Product Availability

Hub Acceptance Rule

The hub must confirm: correct rider, correct product category, correct unit count, correct batch ID, package seal status, product condition, no water damage or tampering. If any item fails, the smart contract locks the batch in INSPECTION_ISSUE state.

D. Field Health Champion Workflow

Purpose: Educate customers, record voluntary payments, and hand over products only after full confirmed payment.

Status	What champion sees	Required action	Next app action
No active customer	'Enroll a customer only with consent'	Enter minimum required details	Add Customer
Customer enrolled	Product option and full price shown	Explain product, safe use, full price, payment choices	Start Purchase Plan
Customer deciding	'Customer has not selected a product'	Wait	View Customer Status
Installment active	Progress card: 0 to 11,400 TZS paid (on-chain verified)	Record only confirmed payments	Check Payment Status
Payment pending	'Do not mark progress as complete — awaiting provider confirmation'	Wait	Refresh Payment Status
Payment confirmed	Progress updates from on-chain event	Explain remaining amount	Record Next Customer Visit
Full payment complete	'Customer has paid in full' (PaymentConfirmed + full amount on-chain)	Prepare approved product	Complete Handover
Handover required	Customer, product, payment total, stock reference	Give product and education	Confirm Handover
Handover complete	'Product handed over. Margin sent via mobile money.' (HandoverCompleted + MarginDistributed on-chain)	Give digital receipt with blockchain link	View Receipt
No payment for 14 days	'Customer paused payments. Do not pressure or auto-charge.'	Send respectful reminder or close plan	Contact Customer or Close Plan
Customer requests refund	'Do not promise a refund outside policy'	Submit request	Request Refund Review

Customer Enrollment Rule

The champion collects only: customer name or preferred name, verified phone number, chosen product category, consent for transaction/reminder messages, optional broad service area.

The champion does NOT collect: menstrual-cycle history, health diagnosis, pregnancy status, sexual history, exact home location, or child information without safeguarding consent.

Customer Payment Rule

A customer may choose: full payment of 11,400 TZS, voluntary installments (e.g., 1,000 TZS at a time), or a different installment amount if allowed.

```
PRODUCT PRICE: 11,400 TZS
CONFIRMED PAID (on-chain): 6,000 TZS
REMAINING: 5,000 TZS

NEXT ACTION:
Make your next voluntary payment.

IMPORTANT:
No automatic deductions. No late fee. No debt.
All payments verified on blockchain.
```

E. Customer Handover Workflow

A product is handed over only when all conditions are true:

- Full product price is confirmed via mobile money provider API (PaymentConfirmed on-chain).
- Correct product is available in champion inventory.
- Product has not been marked damaged, expired, recalled, or unavailable.
- Customer receives basic product-use information.
- Champion confirms physical handover (HandoverCompleted event on-chain).
- Customer receives digital receipt with blockchain explorer link.

For reusable products, the champion confirms the customer understands: washing with water and soap, drying fully before reuse, safe storage, when not to use, when to seek medical care. For disposable products, the champion explains safe use and disposal.

9. Payment Confirmation Rules

Only three payment statuses exist:

Status	Meaning	What user must do
PAYMENT_PENDING	Customer says payment was sent; mobile money provider API has not yet confirmed	Wait; do not release product or complete transfer
PAYMENT_CONFIRMED	Mobile money provider API confirms payment; audit event written to Celo blockchain	Continue to the next physical action
PAYMENT_FAILED_OR_REVIEW	Payment failed, was reversed, cannot be matched, or needs manual review	Contact support; do not complete transfer

Payment Confirmation Requirements

Every confirmed payment record must include:

- Internal transaction reference.
- On-chain audit event hash (blockchain explorer link).
- Mobile money provider transaction reference (from telco API webhook).
- Amount, currency: TZS.
- Sender and recipient phone number hashes.
- Date and time received.
- Payment purpose.
- Linked inventory batch, order, or customer purchase.
- Confirmation status.
- Reconciliation status (auto-verified on-chain).

Critical Rule

A user cannot force a payment confirmation by pressing a button repeatedly, uploading a screenshot, forwarding an SMS, showing a WhatsApp message, or entering a code. Only the mobile money provider's API webhook changes the transaction status. The blockchain audit record provides additional immutable proof.

Payment Verification Flow

Step	Action	Result
1	Customer sends payment via M-Pesa/Airtel/Mixx/HaloPesa	Mobile money transaction initiated
2	Telco API sends webhook to app backend	Payment confirmation received
3	App backend verifies webhook authenticity with telco	Payment confirmed as genuine
4	App backend writes AuditEvent to Celo blockchain	Immutable on-chain record

Step	Action	Result
5	App shows 'Payment confirmed' + 'View on Blockchain' link	User can verify independently
6	Margins sent via mobile money; MarginDistributed recorded on-chain	Stakeholders receive margins + verification

If the blockchain is temporarily unavailable, payments still work normally — the telco API confirmation is the payment confirmation. Audit events are queued and written when connectivity returns.

10. Inventory and Transfer Rules

Required Inventory Information

- Batch ID (also used as on-chain reference identifier).
- Supplier or factory name.
- Product type, reusable or disposable category.
- Unit quantity, date prepared or received.
- Quality status, current custodian (phone hash).
- Current location or service area.
- Seal or package identifier where used.
- Transfer history (on-chain custody chain).

Custody States (with on-chain mapping)

Custody State	On-chain Event	Meaning
AVAILABLE_AT_SUPPLIER	BatchRegistered	Stock ready at supplier
RESERVED_FOR_RIDER	Reservation event	Batch assigned to rider
PAYMENT_PENDING	PaymentConfirmed pending	Rider payment in progress
READY_FOR_PICKUP	PaymentConfirmed on-chain	Stock ready for physical pickup
PICKED_UP	CustodyTransferred	Rider has custody
IN_TRANSIT	In-transit event	Stock being transported
AT_HUB_INSPECTION	Inspection event	Hub inspecting stock
ACCEPTED_AT_HUB	CustodyTransferred	Hub has custody
AVAILABLE_AT_HUB	Available event	Stock available for champion transfer
RESERVED_FOR_CHAMPION	Reservation event	Stock assigned to champion
WITH_CHAMPION	CustodyTransferred	Champion has custody
RESERVED_FOR_CUSTOMER	Reservation event	Product reserved for customer
HANDED_TO_CUSTOMER	HandoverCompleted	Customer received product (final custody)
INSPECTION_ISSUE	ExceptionRaised	Inspection failed; batch locked by smart contract
DAMAGED_OR_QUARANTINED	ExceptionRaised	Stock damaged; cannot be transferred
RETURNED	Return event	Stock returned to previous custodian

Transfer Rule

Ownership or custody changes only after: physical product is inspected, quantity is confirmed, payment is confirmed via telco API where required, both sender and receiver confirm the handover (dual-confirmation on-chain), and the smart contract records the CustodyTransferred event.

If there is an issue, the stock is placed in an exception state on-chain. The smart contract prevents further transfers until an Admin resolves the exception. It is never silently reassigned.

Blockchain creates secure and traceable systems that ensure products reach intended recipients without mismanagement or fraud.¹

1. Blossomflow Foundation, "Technology and Period Poverty." <https://blossomflow.org/technology-and-period-poverty/>

11. Product Safety Rules

Reusable Products

Offer reusable pads or cups only where the customer can access: clean water, soap, privacy, a place to wash, a place to dry products thoroughly, and safe storage.

UNICEF notes that reusable pads must be washed and dried after use.¹

Disposable Products

Offer disposable products where reusables are unsuitable due to water scarcity, lack of privacy, poor washing/drying conditions, or customer preference. The product must meet approved quality and safety requirements. The champion can explain practical disposal options.

Village Product-Pivot Rule

- Hub Manager selects REPORT WASH OR WATER CONCERN (ExceptionRaised on-chain).
- John reviews with local health or community guidance.
- John and USA Co-Founder approve the product availability decision (recorded on-chain).
- The app updates approved products for that location.
- Champions see the updated product options.

1. UNICEF, Menstrual Hygiene Products. <https://www.unicef.org/supply/menstrual-hygiene-products>

12. Problems and Exceptions

Every user sees a large button: REPORT A PROBLEM

Problem	Immediate instruction	App next action
Payment pending too long	Do not release product or stock	Contact Support
Payment was reversed	Stop transfer and preserve evidence	Open Payment Review
Wrong amount paid	Do not adjust manually	Request Payment Review
Stock is short	Count again with sender present	Report Shortage
Product damaged or wet	Separate stock; do not sell	Quarantine Stock
Seal is broken	Do not accept as normal stock	Report Seal Issue
Rider at wrong hub	Do not transfer stock	Contact Admin
Phone lost or SIM changed	Stop use of account	Lock My Account
Customer requests refund	Do not promise a result	Request Refund Review
Customer feels unwell	Do not diagnose	Show Health Referral Guidance
Suspected theft or fraud	Do not accuse publicly	Secure Stock and Contact Admin
No network	Record non-financial note only	Save Offline Note
Blockchain network issue	Wait; payments still work	Audit trail syncing

Offline Rule

The app may allow an offline note for non-financial events. No payment, custody transfer, or handover is completed while offline. No on-chain events are emitted while offline. When connectivity returns, pending audit events are submitted to the blockchain.

Exception Handling On-Chain

When an exception is raised, the smart contract locks the affected batch in INSPECTION_ISSUE or DAMAGED_OR_QUARANTINED state. No further custody transfers can occur until an Admin resolves the exception on-chain (dual-signature required).

13. Admin Dashboard

John and the USA Co-Founder use the Admin Dashboard together. All dashboard data is backed by on-chain events, providing a verifiable audit trail.

Admin Home Screen

TODAY'S PRIORITIES

1. Payments needing review: 2
2. Deliveries awaiting inspection: 1
3. Low-stock hubs: 1
4. New stakeholder requests: 3
5. Customer complaints: 0
6. On-chain exceptions unresolved: 1
7. Audit trail sync status: All synced

NEXT ACTION:

Review payment exceptions

BLOCKCHAIN AUDIT:

[View Full Audit Trail]

Admin Actions

Only Admins can:

- Add, activate, suspend, or remove stakeholders (StakeholderActivated on-chain).
- Approve product and price lists (PriceListUpdated on-chain, dual-signature).
- Approve new hubs, service areas, supplier contracts, and product categories.
- Review payment exceptions (ExceptionResolved on-chain).
- Approve refund outcomes and corrections (recorded on-chain).
- Approve product-category changes due to WASH conditions.
- Review inventory discrepancies (cross-referenced with on-chain custody chain).
- Export approved reports (with on-chain verification links).

Two-Person Approval (dual-signature on-chain)

- Changing a retail price.
- Changing payout or recipient details.
- Activating a new supplier or hub.
- Approving a refund above the pilot threshold.
- Changing a completed financial record.
- Viewing or exporting large customer data sets.
- Adding a new city or village.
- Changing privacy, data-retention, or payment-provider settings.

On-Chain Audit Trail

The Admin Dashboard includes a 'Blockchain Audit' section displaying all on-chain events in chronological order. Each event is clickable and opens full transaction details on the blockchain explorer. Blockchain-based accounting significantly enhances supply chain transparency, reduces commodity leakage, and strengthens governance accountability.¹

1. African Journal of Reproductive Health, blockchain in reproductive health supply chains.
<https://ajrh.org/index.php/ajrh/article/view/7124/2622>

14. Privacy and Data Rules

Minimum data only

Person	Minimum data	On-chain data
Supplier	Business name, contact, location, pricing data	Wallet hash, batch IDs, payment events
Rider	Name, phone, role, service area, payout route	Wallet hash, custody events, margin events
Hub Manager	Name, phone, hub location, storage, payout route	Wallet hash, custody events, margin events
Field Champion	Name, phone, service area, training, payout route	Wallet hash, handover events, margin events
Customer	Name or preferred name, phone, product, payment progress, consent	Wallet hash, payment events, handover receipt

Access Rules

- Riders see only their assigned pickups and deliveries.
- Hub Managers see only their hub stock and approved champion transfers.
- Champions see only their own customers.
- Customers see only their own purchase record (with on-chain verification links).
- Admins see data necessary to operate, audit, and support the system.
- On-chain data is pseudonymous: phone hashes are visible, personal data is off-chain.
- The USA Co-Founder's access must be assessed as a cross-border data-access arrangement under Tanzanian requirements.¹

Privacy Notice

We collect your name, phone number, product choice, and payment status to manage your purchase and send receipts or reminders. Your payments are recorded on a public blockchain for transparency, but your personal identity is not stored on the blockchain — only a pseudonymous wallet address is used. We do not sell your information. We do not collect menstrual-cycle or medical details. You can ask us to correct your information or stop reminders. For help, contact John or the support number.

1. Tanzania PDPC. <https://pdpc.go.tz/>

15. Daily Operating Routine

Morning: John

- Open Admin Dashboard.
- Review payment exceptions (cross-reference with on-chain events).
- Review scheduled factory pickups and expected hub deliveries.
- Confirm low-stock alerts and new stakeholder requests.
- Review on-chain exceptions unresolved.
- Send support messages to anyone blocked in pending status.

Admin next action: Resolve Oldest Pending Item

Morning: Hub Manager

- Open app, check expected rider delivery and stock level.
- Prepare clean, dry inspection and storage area.
- Check approved champion requests.
- Verify overnight on-chain payment confirmations.

Hub next action: View Today's Deliveries

Morning: Field Champion

- Open app, check customer payment confirmations (on-chain verified).
- Check customers who have fully paid.
- Prepare education materials and approved product inventory.
- Contact only customers who consented to reminders.

Champion next action: Review Customer Payments

End of day: John and USA Co-Founder

- Review all pending payments and stock in exception status.
- Confirm every completed transfer has matching on-chain payment and custody events.
- Review complaints, record issues and resolutions.
- Check next day's stock and staff readiness.
- Run automated on-chain reconciliation report.

Admin next action: Complete Daily Reconciliation

Daily reconciliation is automated: the system cross-references on-chain PaymentConfirmed events with CustodyTransferred and HandoverCompleted events. Any mismatch is flagged for Admin review.

16. Weekly Review

Every week, John and the USA Co-Founder review:

- Units purchased from suppliers (on-chain batch events).
- Units received at hubs (CustodyTransferred events).
- Units transferred to champions (CustodyTransferred events).
- Units handed to customers (HandoverCompleted events).
- Confirmed payments vs expected (PaymentConfirmed events).
- Pending payments older than agreed limit.
- Stock shortages or damage (ExceptionRaised events).
- Customer completion rate (full payment + handover on-chain).
- Customer complaints and refund requests.
- Champion, hub, and rider feedback.
- Product suitability: reusable vs disposable.
- On-chain audit trail completeness.
- Whether the pilot remains safe, financially viable, and trusted.

Stop-and-Fix Triggers

Pause expansion and investigate if:

- Any payment cannot be reconciled (on-chain mismatch).
- Suspected fraud event (unusual on-chain activity).
- Stock loss exceeds pilot threshold.
- Customer complaint involves safety, coercion, or privacy.
- Reusable product used where safe washing/drying conditions are unavailable.
- Stakeholder repeatedly ignores no-credit or payment-confirmation rules.
- Mobile money provider API is unavailable or unreliable.
- Personal data is accessed or shared improperly.
- Smart contract exception lock cannot be resolved.

17. First 10 Days

Day 1: Legal, payment, and blockchain setup

- Select Tanzanian legal entity and operating structure.
- Register business with Vodacom for M-Pesa Daraja API access (free).
- Register business with Airtel for Airtel Money API access (free).
- Deploy AuditTrail and CustodyChain smart contracts on Celo testnet.
- Configure Celo SocialConnect for phone-number-to-wallet mapping.
- Begin PDPC registration and privacy compliance.¹

Day 2: Select pilot area

- Choose one city or village.
- Choose one product category or limited catalogue.
- Confirm customer need, safety conditions, transport route, mobile-network reliability.
- Verify M-Pesa and Airtel network coverage in the area.

Day 3: Verify supplier

- Confirm supplier identity, product quality, price, MOQ, lead time, batch labeling, pickup process.
- Obtain written commercial terms.
- Do not publicly call any organization a partner until agreed in writing.

Day 4: Verify hub

- Inspect storage: dry, secure, clean conditions.
- Confirm stock-count process, hub manager's phone, role, and training.

Day 5: Verify one or two riders

- Confirm identity, phone, motorcycle reliability, safe transport ability, route knowledge.
- Train on payment confirmation, custody, seal inspection, and exceptions.
- Verify mobile money account readiness (M-Pesa or Airtel).

Day 6: Verify field champions

- Choose small group of trusted, trained community-facing workers.
- Train on product education, privacy, customer consent, no-credit rule, escalation.
- Explain blockchain transparency: payments verified on-chain, margins recorded.

Day 7: Configure app and deploy smart contracts

- Load approved users, supplier, hub, product and price list.
- Load M-Pesa API integration (Phase 1 provider).
- Deploy smart contracts to Celo mainnet (or testnet if preferred for pilot).

- Test every screen using test data.
- Test on-chain event recording and blockchain explorer verification.
- Test Celo SocialConnect phone-number mapping.

Day 8: Run a dry test

- Use test data and test API sandbox (no real money).
- Test: rider activation, factory pickup, pending payment, payment confirmation via API sandbox, hub inspection, inventory transfer, champion transfer, customer installment, full-payment handover, damaged-stock exception, offline note, lost-phone lock, blockchain explorer verification.

Day 9: First real stock run

- Use small quantity. John available throughout.
- Do not run multiple routes at once. Record all issues.
- All payments via M-Pesa (Phase 1). Audit events on Celo mainnet.

Day 10: Review and decide

- John and USA Co-Founder decide: continue, fix problems, pause, or add one champion if all controls worked.
- Review on-chain audit trail for completeness.
- Verify all custody events match physical movements.

1. Tanzania PDPC. <https://pdpc.go.tz/>

18. Simple App Text Templates

Stakeholder Activation Message

Habari [Name].

You have been approved as a [Role] in the menstrual health supply pilot.

Your account was added manually after verification by John and the USA Co-Founder.
You cannot transfer your account to another person.

Your transactions will be recorded on a public blockchain for full transparency.
You do not need to do anything extra – it happens automatically.

Open the app and complete your first action:
Confirm your phone and create your PIN.

If you need help, contact John directly.

Rider Pickup Message

Habari [Name].

A pickup has been assigned to you.

Product: [Product]
Quantity: [Quantity]
Supplier: [Supplier]
Pickup location: [Location]
Approved total price: [Amount] TZS

Next action:
Open the app, review the pickup, and select Accept Pickup.

Pay via M-Pesa, Airtel Money, Mixx by Yas, or HaloPesa.
Do not collect stock until the app shows:
PAYMENT CONFIRMED.

Customer Purchase Message

Habari [Name].

Your product price is [Amount] TZS.

Confirmed paid (on-chain): [Amount] TZS
Remaining: [Amount] TZS

You may pay voluntarily in installments or pay the full amount at once.
Pay via M-Pesa, Airtel Money, Mixx by Yas, or HaloPesa.
There are no automatic deductions, no late fees, and no debt.
All payments are verified on the blockchain for full transparency.

The product will be handed over after full payment is confirmed.

Customer Completion Message

Habari [Name].

Your full payment has been confirmed.

Next action:

Meet your approved Health Champion to receive your product and use instructions.

Your receipt will be available after handover.

Blockchain receipt link: [explorer URL]

Margin Receipt Message

Habari [Name].

Your margin for this sale has been sent to your mobile money wallet.

Amount: [Amount] TZS

Role: [Role]

Recorded on blockchain: [tx hash]

View on blockchain: [explorer URL]

No action needed. This is an automated notification.

19. Smart Contract Specifications

Two smart contracts handle all on-chain activity. No payment settlement occurs on-chain — the contracts only record events. All smart contracts should be audited by a qualified third party before deployment.

19.1 Contract: AuditTrail

Purpose: Records system events on the Celo blockchain for transparency.

Key functions:

- `recordEvent()`: Called by app backend after any confirmed event. Records event type, relevant data (amounts, batch IDs, phone hashes), timestamp. Emits `AuditEvent`.
- `getEvent()`: Publicly readable. Returns full details of any recorded event by ID.
- `getEventsByBatch()`: Publicly readable. Returns all events for a specific product batch (full custody chain).
- `getEventsByStakeholder()`: Publicly readable. Returns all events for a specific phone hash.
- `getDailyReconciliation()`: Publicly readable. Returns daily summary (total payments, transfers, handovers, exceptions).

Access control: Only app backend address can call `recordEvent()`. All read functions are public.

19.2 Contract: CustodyChain

Purpose: Tracks product batch custody from supplier to customer on-chain.

- `registerBatch()`: Called by app backend when a new batch enters the system. Assigns batch ID, product type, quantity, supplier phone hash. Emits `BatchRegistered`.
- `transferCustody()`: Called by app backend when stock moves between stakeholders. Requires both sender and receiver phone hashes. Checks `PaymentConfirmed` exists. Emits `CustodyTransferred`.
- `raiseException()`: Called by app backend when a problem is reported. Locks batch in exception state. Emits `ExceptionRaised`.
- `resolveException()`: Only callable by Admin (dual-signature). Unlocks or redirects batch. Emits `ExceptionResolved`.
- `getBatchHistory()`: Publicly readable. Returns complete custody chain for a batch.

Access control: `transferCustody` and `raiseException` called by app backend. `resolveException` requires dual-admin signature.

19.3 Deployment Requirements

- Deploy on Celo mainnet (or testnet during pilot phase).
- All contracts verified on blockchain explorer (source code published).
- Admin wallet addresses use multi-signature wallets (2-of-2 for John + USA Co-Founder).
- Formal smart contract security audit completed before mainnet deployment.
- Emergency pause functionality available to Admins.

19.4 Gas Cost Estimates

Operation	Gas (Celo)	Cost (USD)	Who pays
AuditEvent (payment confirmed)	~40,000	~\$0.001	Platform budget
CustodyTransferred	~35,000	~\$0.001	Platform budget
HandoverCompleted	~30,000	~\$0.001	Platform budget
ExceptionRaised	~25,000	~\$0.001	Platform budget
Daily reconciliation summary	~50,000	~\$0.001	Platform budget
Monthly total (pilot scale)	~2M	~\$0.04	Platform budget

Total monthly blockchain cost: under \$1 USD. No stakeholder pays any blockchain fee. No stablecoin, no bridge fees, no conversion charges.

20. Final Launch Checklist

Do not launch until every item below is complete.

Business and Legal

- Tanzanian operating entity and roles confirmed.
- Supplier agreement confirmed in writing.
- Product prices and margins approved and stored on-chain.
- M-Pesa Daraja API integration tested (free, no licensing fee).
- Airtel Money API integration tested (free, Phase 2).
- App does not custody money, create wallets, or auto-debit users.
- No stablecoin or cryptocurrency used for any payment.
- Privacy and PDPC compliance completed with qualified local counsel.¹
- Customer privacy notice ready in Swahili and English.
- Refund, complaint, and dispute process ready.

People

- John and USA Co-Founder have Admin access with multi-signature wallet.
- Every stakeholder manually verified.
- Every stakeholder's phone number mapped to Celo wallet via SocialConnect.
- Every stakeholder completed training.
- Every stakeholder understands no credit and no unverified payments.
- Emergency contacts and escalation routes tested.

App

- Every role sees only one next action.
- Payment statuses: pending, confirmed, or review only.
- No client-side button can mark money as confirmed (only telco API webhook).
- Inventory handovers require inspection and on-chain dual-confirmation.
- Exceptions can be reported easily (and lock batches on-chain).
- App can lock a lost phone account.
- Customer data limited, access-controlled, off-chain.
- Daily reconciliation report works (automated on-chain matching).
- Blockchain explorer links work for all transaction types.
- Smart contracts enforce payment-before-release rule.
- Celo SocialConnect phone-number mapping tested.
- Mobile money provider API webhooks tested and verified.

Security (from Red Team Analysis)

- Webhook verification implemented (signature + API callback + duplicate check).
- Rate limiting on webhook endpoint (max 10 per phone per minute).
- IP allowlisting on webhook endpoint (telco IP ranges only).
- PIN lockout policy: 4-digit, 5 max attempts, 30-min lockout, admin reactivation.
- Session timeout: 15-minute inactivity auto-logout.
- Backend wallet private key stored in HSM or secure enclave.
- Multi-signature admin wallet (2-of-2) deployed.
- Emergency pause function tested.
- Customer phone SMS verification at enrollment.
- Offline queue: disk-persisted, append-only, tamper-evident.
- Admin action logging implemented.
- Incident response plan documented.
- Data retention policy: 7 years off-chain, permanent on-chain.
- Receipts generated server-side only.

Pilot Readiness

- One supplier, one hub, one or two riders ready.
- Small number of field champions ready.
- Initial stock quantity small and controlled.
- First customer group informed and voluntary.
- John available during first live run.
- M-Pesa API live (Phase 1). Airtel API ready for Phase 2.
- No expansion planned until first pilot cycle reviewed on-chain.

1. Tanzania PDPC. <https://pdpc.go.tz/>

21. Final Operating Principle

The first goal is not to build a large network. The first goal is to prove one trustworthy local loop with full on-chain transparency:

```
Approved Supplier
→ Verified Rider
→ Verified Hub
→ Trained Health Champion
→ Consenting Customer
→ Confirmed Mobile Money Payment (telco API)
→ Audit Event Recorded on Celo Blockchain
→ Safe Product Handover (HandoverCompleted on-chain)
→ Automated On-Chain Reconciliation
```

If a payment is not confirmed by the mobile money provider's API, wait.

If a product is damaged, stop (ExceptionRaised locks the batch on-chain).

If a person is not verified, do not activate them.

If a customer has not fully paid, do not hand over the product unless a documented subsidy covers it.

If the system becomes confusing, simplify the screen until the next action is obvious.

The blockchain transparency layer ensures that every transaction is verifiable, immutable, and free. No stablecoin, no cryptocurrency, no third-party bridge provider. Participants pay with standard mobile money (M-Pesa, Airtel, Mixx by Yas, HaloPesa) through free direct API integration. The Celo blockchain records what happened as an immutable audit trail. This is how John and the USA Co-Founder establish the right first stakeholders, build trust in the first city or village, and create a reliable foundation for later expansion.

Blockchain-based accounting significantly enhances supply chain transparency, reduces commodity leakage, and strengthens governance accountability in reproductive health service delivery.¹ UNFPA and WFP are already using blockchain to ensure women and girls can access menstrual products reliably, privately, and without standing in long distribution lines.²

1. African Journal of Reproductive Health. <https://ajrh.org/index.php/ajrh/article/view/7124/2622>

2. UNFPA, "In focus: Innovation for menstrual hygiene." <https://asiapacific.unfpa.org/en/innovation-dhaka-blockchain>

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This document was prepared as a pilot operating handbook and app specification. It does not constitute legal advice. All regulatory compliance, licensing, and data protection requirements must be confirmed with qualified Tanzanian legal counsel before launch.